Forecasting with Machine Learning
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Maths Brush-Up

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1 Preliminaries: Sums and Notation

1.1 Sum and Product Notation

The sum and product of $x_1, \ldots, x_n$ are,

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n, \qquad \prod_{i=1}^{n} x_i = x_1 x_2 \cdots x_n \tag{1}$$

The index is a placeholder: $\sum_{i=1}^{n} x_i = \sum_{k=1}^{n} x_k$. Indices can be shifted without changing the sum, for instance

$$\sum_{i=1}^{n} x_i = \sum_{j=0}^{n-1} x_{j+1} \tag{2}$$

Two basic rules:

1. $\sum_{i=1}^{n} c = nc$
2. $\sum_{i=1}^{n}(ax_i + by_i) = a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} y_i \quad (a, b \in \mathbb{R})$

when it says that $(a, b \in \mathbb{R})$ it means that a and b are constants.

A constant can be pulled out of a sum, but a quantity that depends on i cannot. In general $\sum_i x_i y_i \neq (\sum_i x_i)\,(\sum_i y_i)$.

1.2 Double Sums

For a finite array a_{ij} $(i = 1, \ldots, m,\ j = 1, \ldots, n)$ the order of summation does not matter:

$$\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} = \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij} \tag{3}$$

Example: $a_{ij} = i + j$, $m = 2$, $n = 3$: summing over j first gives $(2 + 3 + 4) + (3 + 4 + 5) = 21$, and summing over i first gives $(2 + 3) + (3 + 4) + (4 + 5) = 21$.

When the summand factors, the double sum factors (Difference from Section 1.1: here there are two indices, i and j, in Section 1.1 there was only one, i.

$$\sum_{i=1}^{m}\sum_{j=1}^{n} a_i b_j = \left(\sum_{i=1}^{m} a_i\right)\left(\sum_{j=1}^{n} b_j\right). \tag{4}$$

Squaring a sum produces a double sum:

$$\left(\sum_{i=1}^{n} x_i\right)^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j = \sum_{i=1}^{n} x_i^2 + \sum_{i\neq j} x_i x_j \tag{5}$$

That is, the sum of the squared elements of the identical indices, and the sum of the cross-products when the indices differ.

1.3 Sums as Matrix Expressions

Let $\mathbf{x} = (x_1, \ldots, x_n)^\intercal$ and $\mathbf{y} = (y_1, \ldots, y_n)^\intercal$ be $n \times 1$ column vectors. Then $\mathbf{x}^\intercal$ is a $1 \times n$ row vector, and the product $\mathbf{x}^\intercal\mathbf{y}$ is a $(1 \times n)(n \times 1)$ product, which is 1×1, i.e. a single number. It is computed by multiplying the entries pairwise and adding:

$$\mathbf{x}^\intercal\mathbf{y} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = x_1y_1 + x_2y_2 + \cdots + x_ny_n = \sum_{i=1}^{n} x_iy_i \tag{6}$$

It holds that,

$$\sum_{i=1}^{n} x_iy_i = \mathbf{x}^\intercal\mathbf{y}, \qquad \sum_{i=1}^{n} x_i^2 = \mathbf{x}^\intercal\mathbf{x}, \qquad \sum_{i=1}^{n} x_i = \mathbf{1}^\intercal\mathbf{x} \tag{7}$$

Example: Let $\mathbf{x} = (1, 2, 3)^\intercal$ and $\mathbf{y} = (4, 0, -1)^\intercal$ ($n = 3$),

$$\mathbf{x}^\intercal\mathbf{y} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix} = 1 \cdot 4 + 2 \cdot 0 + 3 \cdot (-1) = 4 + 0 - 3 = 1, \tag{8}$$

$$\mathbf{x}^\intercal\mathbf{x} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 = 1 + 4 + 9 = 14, \tag{9}$$

$$\mathbf{1}^\intercal\mathbf{x} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \cdot 1 + 1 \cdot 2 + 1 \cdot 3 = 1 + 2 + 3 = 6 \tag{10}$$

Equivalent with the brute-force sums,

$$\sum_{i=1}^{n} x_iy_i = 1 \cdot 4 + 2 \cdot 0 + 3 \cdot (-1) = 1, \qquad \sum_{i=1}^{n} x_i^2 = 14, \qquad \sum_{i=1}^{n} x_i = 6 \tag{11}$$

The trace is the sum of the diagonal elements of the matrix (see Section 3.5 for a full exposition),

$$\mathrm{tr}(\mathbf{A}) = \sum_{i=1}^{n} a_{ii} \tag{12}$$

Example: Let $\mathbf{A} = \begin{bmatrix} \mathbf{1} & 4 & 7 \\ 3 & \mathbf{4} & 1 \\ 8 & 3 & \mathbf{9} \end{bmatrix}$. Then,

$$\mathrm{tr}(\mathbf{A}) = \mathrm{tr} \begin{bmatrix} \mathbf{1} & 4 & 7 \\ 3 & \mathbf{4} & 1 \\ 8 & 3 & \mathbf{9} \end{bmatrix} = \sum_{i=1}^{n} a_{ii} = 1 + 4 + 9 = 14 \tag{13}$$

1.4 Summing Rows

Let $\mathbf{x}_i$ denote the i-th row of the $n \times k$ matrix $\mathbf{X}$ (a $1 \times k$ vector) and y_i the i-th entry of $\mathbf{y}$, then

$$\mathbf{X}^\intercal\mathbf{X} = \sum_{i=1}^{n} \mathbf{x}_i^\intercal \mathbf{x}_i, \qquad \mathbf{X}^\intercal\mathbf{y} = \sum_{i=1}^{n} \mathbf{x}_i^\intercal y_i \tag{14}$$

Example: Let $\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, with rows $\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (3, 4)$. Then,

$$\mathbf{x}_1^\intercal\mathbf{x}_1 + \mathbf{x}_2^\intercal\mathbf{x}_2 = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix} = \mathbf{X}^\intercal\mathbf{X} \tag{15}$$

$$\mathbf{x}_1^\intercal y_1 + \mathbf{x}_2^\intercal y_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \mathbf{X}^\intercal\mathbf{y} \tag{16}$$

1.5 Dimensional Alignment

The product of an $m \times n$ matrix and an $n \times p$ matrix exists and is $m \times p$ – the inner dimensions must be equivalent for the product to exist. In the case of matrix multiplication, if this holds, the matrices are said to be conformable. For $\mathbf{x}, \mathbf{y}$ of dimension $n \times 1$ and $\mathbf{X}$ of dimension $n \times k$:

$$\begin{array}{lll} \mathbf{x}^\intercal\mathbf{y}: & (1 \times n)(n \times 1) = 1 \times 1 & \text{(a scalar)} \\ \mathbf{x}\mathbf{y}^\intercal: & (n \times 1)(1 \times n) = n \times n & \text{(a matrix)} \\ \mathbf{X}^\intercal\mathbf{X}: & (k \times n)(n \times k) = k \times k & \text{(a matrix)} \\ \mathbf{X}^\intercal\mathbf{y}: & (k \times n)(n \times 1) = k \times 1 & \text{(a vector)} \\ \mathbf{X}\mathbf{y}: & (n \times k)(n \times 1) & \text{undefined unless } k = n \end{array} \tag{17}$$

Example: If $\mathbf{A} \in \mathbb{R}^{3\times2}$, $\mathbf{B} \in \mathbb{R}^{2\times4}$, and $\mathbf{C} \in \mathbb{R}^{4\times1}$, then $\mathbf{ABC}$ exists (and is $\mathbf{ABC} \in \mathbb{R}^{3\times1}$),

$$(3 \times 2) \times (2 \times 4) \times (4 \times 1) = (3 \times 4) \times (4 \times 1) = 3 \times 1 \tag{18}$$

$\mathbf{A}^\intercal\mathbf{B}$ does not exist because $\mathbf{A}^\intercal \in \mathbb{R}^{2\times3}$ and $\mathbf{B} \in \mathbb{R}^{2\times4}$.

2 Definitions, Linear Algebra

2.1 Matrix

An $m \times n$ matrix $\mathbf{A} = [a_{ij}]$ is a (2D) rectangular array with m rows and n columns. a_{ij} denotes the generic entry in row i, column j,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \tag{19}$$

Example:

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 7 \\ -4 & 5 & 0 \end{bmatrix}, \qquad a_{13} = 7. \tag{1}$$

Square matrix: A matrix with $m = n$.

Vectors:

Let $\mathbf{x} \in \mathbb{R}^{1\times m}$. This is a special case of a matrix, a row vector,

$$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \cdots & y_n \end{bmatrix} \tag{20}$$

Let $\mathbf{y} \in \mathbb{R}^{n\times 1}$. This is the special case of a matrix which is a column vector,

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \tag{21}$$

Diagonal matrix: A diagonal matrix is square matrix with $a_{ij} = 0$ for $i \neq j$:

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix} \tag{22}$$

Identity matrix: The identity $\mathbf{I}_n$ is the diagonal matrix with ones on the diagonal:

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \tag{23}$$

Zero matrix: The zero matrix $\mathbf{0}$ has every entry equal to zero and need not be square.

3 Matrix Operations

3.1 Addition

Matrices of the same dimension are added entrywise: $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 2 & -1 & 7 \\ -4 & 5 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -4 \\ 4 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 3 \\ 0 & 7 & 3 \end{bmatrix} \tag{24}$$

Matrices of different dimensions cannot be added.

3.2 Scalar Multiplication

For a real number γ (this we write $\gamma \in \mathbb{R}$), $\gamma\mathbf{A} = [\gamma a_{ij}]$. That is, multiplying a matrix with a constant is equivalent to multiplying each element with the constant.

Example: Let $\mathbf{A} = \begin{bmatrix} 1 & 4 & 7 \\ 3 & 4 & 1 \\ 8 & 3 & 9 \end{bmatrix}$ and $\gamma = 2$. Then,

$$\gamma\mathbf{A} = \begin{bmatrix} 2 & 8 & 14 \\ 6 & 8 & 2 \\ 16 & 6 & 18 \end{bmatrix} \tag{25}$$

3.3 Matrix Multiplication

The product $\mathbf{AB}$ exists when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$ (that is, the matrices are conformable, see Section 1.5). If $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$, then $\mathbf{AB} \in \mathbb{R}^{m \times p}$ with,

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \tag{26}$$

That is, entry (i, j) is the row i of $\mathbf{A}$ multiplied term by term with column j of $\mathbf{B}$ and summed.

Example:

$$\begin{bmatrix} 2 & -1 & 0 \\ -4 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 6 & 0 \\ -1 & 2 & 0 & 1 \\ 3 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 12 & -1 \\ -1 & -2 & -24 & 1 \end{bmatrix} \tag{27}$$

If $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{y} \in \mathbb{R}^{m \times 1}$, then $\mathbf{Ay} \in \mathbb{R}^{n \times 1}$. If $\mathbf{x} \in \mathbb{R}^{1 \times n}$, then $\mathbf{xA} \in \mathbb{R}^{1 \times m}$.

The matrix multiplication operation does not commute (see rule 14). This means that below you will find no rule such as $AB = BA$. Matrix multiplication always occurs from left to right (think: direction we read). When manipulating matrices (linear algebra), we can talk about pre-multiplying a matrix (for a matrix A, we pre-multiply B: BA) or post-multiplying a matrix (for a matrix A, we post-multiply B: AB). The distinction is crucial, because the resulting matrices depend heavily on the order of multiplication.

Matrix operations For conformable matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and scalars α, β:

1. $(\alpha + \beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{A}$
2. $\alpha(\mathbf{A} + \mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}$
3. $(\alpha\beta)\mathbf{A} = \alpha(\beta\mathbf{A})$
4. $\alpha(\mathbf{AB}) = (\alpha\mathbf{A})\mathbf{B}$
5. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
6. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
7. $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

8. $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}$
9. $(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}$
10. $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$
11. $\mathbf{A}+\mathbf{0} = \mathbf{0}+\mathbf{A} = \mathbf{A}$
12. $\mathbf{A}-\mathbf{A} = \mathbf{0}$
13. $\mathbf{A0} = \mathbf{0A} = \mathbf{0}$
14. In general $\mathbf{AB} \neq \mathbf{BA}$, even when both products are defined.

If $\mathbf{A} \in \mathbb{R}^{n\times m}$ and $\mathbf{B} \in \mathbb{R}^{m\times p}$, then $\mathbf{BA}$ is defined only when $n = p$.
Symmetric matrix: A square matrix with $\mathbf{A}^\intercal = \mathbf{A}$. For any matrix $\mathbf{X} \in \mathbb{R}^{n\times k}$, the product $\mathbf{X}^\intercal\mathbf{X}$ is symmetric.

3.4 Transpose

The transpose $\mathbf{A}^\intercal$ of an $m \times n$ matrix $\mathbf{A} = [a_{ij}]$ is the $n \times m$ matrix $[a_{ji}]$, obtained by swapping rows and columns:

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 7 \\ -4 & 5 & 0 \end{bmatrix}, \qquad \mathbf{A}^\intercal = \begin{bmatrix} 2 & -4 \\ -1 & 5 \\ 7 & 0 \end{bmatrix} \tag{28}$$

Properties:

1. $(\mathbf{A}^\intercal)^\intercal = \mathbf{A}$
2. $(\alpha\mathbf{A})^\intercal = \alpha\mathbf{A}^\intercal$
3. $(\mathbf{A}+\mathbf{B})^\intercal = \mathbf{A}^\intercal + \mathbf{B}^\intercal$
4. $(\mathbf{AB})^\intercal = \mathbf{B}^\intercal\mathbf{A}^\intercal$ ($\mathbf{A} \in \mathbb{R}^{m\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times k}$)
5. $\mathbf{x}^\intercal\mathbf{x} = \sum_{i=1}^{n} x_i^2$ for a vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$
6. If $\mathbf{A} \in \mathbb{R}^{n\times k}$ with rows $\mathbf{a}_1, \ldots, \mathbf{a}_n$ (each $\in \mathbb{R}^{1\times k}$), i.e. $\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_n \end{bmatrix}$, then $\mathbf{A}^\intercal = (\mathbf{a}_1^\intercal \;\; \mathbf{a}_2^\intercal \;\; \cdots \;\; \mathbf{a}_n^\intercal)$

3.5 Trace

For a square matrix, $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\operatorname{tr}(\mathbf{A}) = \sum_{i=1}^{n} a_{ii}$. Properties:

1. $\operatorname{tr}(\mathbf{I}_n) = n$
2. $\operatorname{tr}(\mathbf{A}^\intercal) = \operatorname{tr}(\mathbf{A})$
3. $\operatorname{tr}(\mathbf{A}+\mathbf{B}) = \operatorname{tr}(\mathbf{A}) + \operatorname{tr}(\mathbf{B})$
4. $\operatorname{tr}(\alpha\mathbf{A}) = \alpha\operatorname{tr}(\mathbf{A})$
5. $\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$

3.6 Matrix Inverse

A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible (nonsingular) if there exists a matrix $\mathbf{A}^{-1}$ with,

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n = \mathbf{A}\mathbf{A}^{-1} \tag{29}$$

Otherwise it is singular.
Properties:

1. The inverse, if it exists, is unique
2. $(\alpha\mathbf{A})^{-1} = (1/\alpha)\mathbf{A}^{-1}$ for $\alpha \neq 0$
3. $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ for invertible matrices, $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$
4. $(\mathbf{A}^\intercal)^{-1} = (\mathbf{A}^{-1})^\intercal$

4 Linear Independence and Rank

Vectors $\mathbf{x}_1, \ldots, \mathbf{x}_r$ (each $\in \mathbb{R}^{n \times 1}$) are *linearly independent* if

$$\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \cdots + \alpha_r\mathbf{x}_r = \mathbf{0} \tag{2}$$

holds only when $\alpha_1 = \cdots = \alpha_r = 0$. If Eq. (2) holds with some $\alpha_j \neq 0$, the vectors are linearly dependent, which is the same as saying at least one of them is a linear combination of the others.

The *rank* of a matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\text{rank}(\mathbf{A})$, is the maximal number of linearly independent columns. If $\text{rank}(\mathbf{A}) = m$, then $\mathbf{A}$ has full column rank.
Example:

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 0 & 0 \end{bmatrix} \tag{30}$$

has rank 1, since the second column is 3 times the first.
Properties:

1. $\text{rank}(\mathbf{A}^\intercal) = \text{rank}(\mathbf{A})$
2. If $\mathbf{A} \in \mathbb{R}^{n \times k}$, then $\text{rank}(\mathbf{A}) \leq \min\{n, k\}$
3. If $\mathbf{A} \in \mathbb{R}^{k \times k}$ and $\text{rank}(\mathbf{A}) = k$, then $\mathbf{A}$ is invertible

5 Random Vectors

5.1 Expected value

For a random vector $\mathbf{y} \in \mathbb{R}^{n \times 1}$, $\mathbf{y} = [y_1, ..., y_n]^\intercal$, it holds that $\mathbf{y}$, $\mathbb{E}[\mathbf{y}] = [\mathbb{E}[y_1], \ldots, \mathbb{E}[y_n]]^\intercal$. That is, the expected value of the vector is equal to the expected value of the elements of the vector.

For a random matrix $\mathbf{Z}$, $\mathbb{E}[\mathbf{Z}] = [\mathbb{E}[z_{ij}]]$. That is, the expected value of the matrix is equal to the expected value of its generic elements.
For nonrandom $\mathbf{A}$, $\mathbf{B}$, $\mathbf{b}$ of conformable sizes:

$$\mathbb{E}[\mathbf{A}\mathbf{y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{y}] + \mathbf{b}, \qquad \mathbb{E}[\mathbf{A}\mathbf{Z}\mathbf{B}] = \mathbf{A}\mathbb{E}[\mathbf{Z}]\mathbf{B} \tag{31}$$

This means that the constant matrix $\mathbf{A}$ and constant vector $\mathbf{b}$ can be taken outside the expectation operator for the random vector case. The same holds for the constant matrices $\mathbf{A}$ and $\mathbf{B}$ in the random matrix case.

5.2 Covariance matrix

The covariance matrix of $\mathbf{y}$ is defined as,

$$\mathbb{V}[\mathbf{y}] = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{bmatrix} \tag{32}$$

with $\sigma_j^2 = \mathbb{V}[y_j]$ and $\sigma_{ij} = \text{Cov}[y_i, y_j]$. It is symmetric because $\text{Cov}[y_i, y_j] = \text{Cov}[y_j, y_i]$. Covariances are contained on the off-diagonal, while the diagonal holds the variances.
Properties:
For a single variable, $\mathbb{V}[aX + b] = a^2\mathbb{V}[X]$: the constant b only shifts the values, and a stretches the deviations from the mean, which enter variance squared. For a random vector $\mathbf{y}$ the same rule applies,

1. If the elements of $\mathbf{y}$ are uncorrelated, $\mathbb{V}[\mathbf{y}]$ is a diagonal matrix. If they also all have variance σ^2 (the same variance for all instances of $\mathbf{y} = [y_1, ..., y_n]^\intercal$), then $\mathbb{V}[\mathbf{y}] = \sigma^2\mathbf{I}_n$
2. For nonrandom $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^{m\times 1}$ it holds that $\mathbb{V}[\mathbf{A}\mathbf{y} + \mathbf{b}] = \mathbf{A}\mathbb{V}[\mathbf{y}]\mathbf{A}^\intercal$
3. For a weighted sum with fixed weights $\mathbf{a}$, taking $\mathbf{A} = \mathbf{a}^\intercal$: $\mathbb{V}[\mathbf{a}^\intercal\mathbf{y}] = \mathbf{a}^\intercal\mathbb{V}[\mathbf{y}]\mathbf{a} \geq 0$, since a variance is never negative

5.3 Normal distribution

A random variable X has a normal (Gaussian) distribution with mean μ and variance σ^2, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if its density (probability density function) is,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{33}$$

Two important properties,

1. If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a, b \in \mathbb{R}$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$
2. When $\mu = 0, \sigma^2 = 1$, the distribution is called the standard normal distribution.

5.4 Multivariate normal distribution

Let $\mathbf{y} \in \mathbb{R}^{n\times 1}$ be a random vector with mean vector $\boldsymbol{\mu} = \mathbb{E}[\mathbf{y}]$ and covariance matrix $\boldsymbol{\Sigma} = \mathbb{V}[\mathbf{y}]$. The vector $\mathbf{y}$ is multivariate normal if every weighted sum of its elements is normally distributed, and it is then determined entirely by $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ – these are the only parameters necessary to fully describe their distribution. We write $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Properties:

1. Each element of $\mathbf{y}$ is normally distributed
2. Two elements y_i and y_j are independent if and only if they are uncorrelated, i.e., $\sigma_{ij} = \text{Cov}[y_i, y_j] = 0$
3. $\mathbf{A}\mathbf{y} + \mathbf{b} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\intercal)$ for nonrandom $\mathbf{A}$ and $\mathbf{b}$. This is the vector version of Property 1 above

6 Vectors, Inner Products and Norms

6.1 Inner product and orthogonality

For $n \times 1$ vectors, the inner product is $\mathbf{x}^\intercal\mathbf{y} = \sum_{i=1}^n x_i y_i$. The inner product is symmetric ($\mathbf{x}^\intercal\mathbf{y} = \mathbf{y}^\intercal\mathbf{x}$). The length (Euclidean norm) of $\mathbf{x}$ is $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\intercal\mathbf{x}} = \sqrt{\sum_{i=1}^n x_i^2}$. Two vectors are *orthogonal* if $\mathbf{x}^\intercal\mathbf{y} = 0$, which means that they are perpendicular.

Example: $(1, 2)^\intercal$ and $(2, -1)^\intercal$ are orthogonal since $2 - 2 = 0$.

6.2 Norms

For a vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$ and a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$,

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|, \qquad \|\mathbf{x}\|_2 = \left(\sum_{i=1}^n x_i^2\right)^{1/2}, \qquad \|\mathbf{A}\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2\right)^{1/2} = \sqrt{\text{tr}(\mathbf{A}^\intercal\mathbf{A})} \quad (34)$$

the ℓ_1, ℓ_2, and Frobenius norms, respectively. The former two are also known as the Manhattan and Euclidean norms.

Properties (each satisfies the properties for norms)[1]:

1. $\|\mathbf{x}\| \geq 0$ with equality only for $\mathbf{x} = \mathbf{0}$
2. $\|c\mathbf{x}\| = |c| \ \|\mathbf{x}\|$
3. Triangle inequality
$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| \quad (35)$$

[1] For matrices $\mathbf{A}, \mathbf{B}$ of equivalent same size, the Frobenius norm satisfies:

1. $\|\mathbf{A}\|_F \geq 0$ with equality only for $\mathbf{A} = \mathbf{0}$
2. $\|c\mathbf{A}\|_F = |c| \ \|\mathbf{A}\|_F$
3. Triangle inequality: $\|\mathbf{A} + \mathbf{B}\|_F \leq \|\mathbf{A}\|_F + \|\mathbf{B}\|_F$

Example: for $\mathbf{x} = (3, -4)^\intercal$, $\|\mathbf{x}\|_1 = 7$ and $\|\mathbf{x}\|_2 = 5$.

7 Mean Squared Error

Let $\hat{\theta}$ be a random variable and θ a constant. Recall that $\mathbb{V}\left[\hat{\theta}\right] = \mathbb{E}\left[\left(\hat{\theta} - \mathbb{E}\left[\hat{\theta}\right]\right)^2\right]$. Then

$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \mathbb{V}\left[\hat{\theta}\right] + \left(\mathbb{E}\left[\hat{\theta}\right] - \theta\right)^2 \tag{36}$$

i.e., variance plus squared bias. This exemplified the bias-variance trade-off.
Example: Let $\theta = 1$, and let $\hat{\theta}$ be a discrete random variable which takes on the value 1 with probability 0.5 and the value 3 with probability 0.5.
The mean and variance of $\hat{\theta}$ are,

$$\mathbb{E}\left[\hat{\theta}\right] = 1 \times 0.5 + 3 \times 0.5 = 2, \qquad \mathbb{E}\left[\hat{\theta}^2\right] = 1^2 \times 0.5 + 3^2 \times 0.5 = 5, \tag{37}$$

$$\mathbb{V}\left[\hat{\theta}\right] = \mathbb{E}\left[\hat{\theta}^2\right] - \left(\mathbb{E}\left[\hat{\theta}\right]\right)^2 = 5 - 2^2 = 1 \tag{38}$$

and the bias is

$$\mathbb{E}\left[\hat{\theta}\right] - \theta = 2 - 1 = 1 \tag{39}$$

so the right-hand side of the decomposition is $\mathbb{V}\left[\hat{\theta}\right] + \left(\mathbb{E}\left[\hat{\theta}\right] - \theta\right)^2 = 1 + 1^2 = 2$.
For the left-hand side, $(\hat{\theta} - \theta)^2$ equals $(1-1)^2 = 0$ when $\hat{\theta} = 1$ and $(3-1)^2 = 4$ when $\hat{\theta} = 3$, so

$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = 0 \cdot 0.5 + 4 \cdot 0.5 = 2 \tag{40}$$

Both sides equal 2.

Now suppose we want to predict a new outcome $Y = \theta + \varepsilon$, where θ is the predictable part and ε is noise with $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{V}[\varepsilon] = \sigma^2$, independent of $\hat{\theta}$ (computed from other data).
Writing $Y - \hat{\theta} = \varepsilon + (\theta - \hat{\theta})$ and expanding the square, the cross term has expectation zero because $\mathbb{E}[\varepsilon] = 0$ and ε is independent of $\hat{\theta}$, so,

$$\mathbb{E}\left[(Y - \hat{\theta})^2\right] = \underbrace{\sigma^2}_{\text{irreducible error}} + \underbrace{\mathbb{V}\left[\hat{\theta}\right]}_{\text{variance}} + \underbrace{\left(\mathbb{E}\left[\hat{\theta}\right] - \theta\right)^2}_{\text{squared bias}} \tag{41}$$

The variance and the bias can be reduced by choosing $\hat{\theta}$ better, but σ^2 cannot, since no choice of $\hat{\theta}$ can predict the noise ε.